ZIYANG CHEN

Ann Arbor, MI ◇ czyang@umich.edu ◇ Webpage

EDUCATION

University of Michigan, Ann Arbor Jan. 2022 - 2025 (Expected)
Ph.D. candidate in Electrical & Computer Engineering | Advisor: *Andrew Owens*

University of Michigan, Ann Arbor Sept. 2019 - May 2021
Master of Science in Engineering, Computer Vision
Overall GPA: **4.0/4.0** | Advisor: *Andrew Owens*

Shanghai Jiao Tong University, UM-SJTU JI Sept. 2015 - Aug. 2019
Bachelor of Science in Engineering
Major: Electrical & Computer Engineering | Minor: Entrepreneurship
GPA: **3.60/4.0** with honors & *Graduated magna cum laude*

PUBLICATIONS

(* indicates equal contribution)

- *Ziyang Chen, Prem Seetharaman, Bryan Russell, Oriol Nieto, David Bourgin, Andrew Owens and Justin Salamon* "Video-Guided Foley Sound Generation with Multimodal Controls". (arXiv 2024) (Project site)
- *Chao Feng, Ziyang Chen, Aleksander Holynski, Alexei A Efros, and Andrew Owens.* "GPS as a Control Signal for Image Generation". (arXiv 2025) (Project site)
- *Ziyang Chen, Daniel Geng and Andrew Owens.* "Images that Sound: Composing Images and Sounds on a Single Canvas". (NeurIPS 2024) (Project site)
- *Ziyang Chen, Israel D. Gebru, Christian Richardt, Anurag Kumar, William Laney, Andrew Owens, and Alexander Richard.* "Real Acoustic Fields: An Audio-Visual Room Acoustics Dataset and Benchmark". (CVPR 2024, **Highlight**) (Project site)
- *Fengyu Yang*, Chao Feng*, Ziyang Chen*, Hyoungseob Park, Daniel Wang, Yiming Dou, Ziyao Zeng, Xien Chen, Rit Gangopadhyay, Andrew Owens, and Alex Wong.* "Binding Touch to Everything: Learning Unified Multimodal Tactile Representations". (CVPR 2024) (Project site)
- *Ziyang Chen, Shengyi Qian and Andrew Owens.* "Sound Localization from Motion: Jointly Learning Sound Direction and Camera Rotation". (ICCV 2023) (Project site)
- *Yuexi Du, Ziyang Chen, Justin Salamon, Bryan Russell, and Andrew Owens.* "Conditional Generation of Audio from Video via Foley Analogies". (CVPR 2023)(Project site)
- *Chao Feng, Ziyang Chen and Andrew Owens.* "Self-Supervised Video Forensics by Audio-Visual Anomaly Detection". (CVPR 2023, **Highlight**) (Project site)
- *Ziyang Chen, David F. Fouhey and Andrew Owens.* "Sound Localization by Self-Supervised Time Delay Estimation". (ECCV 2022) (Project site)
- *Xixi Hu*, Ziyang Chen* and Andrew Owens.* "Mix and Localize: Localizing Sound Sources in Mixtures". (CVPR 2022) (Project site)
- *Ziyang Chen*, Xixi Hu* and Andrew Owens.* "Structure from Silence: Learning Scene Structure from Ambient Sound". (CoRL 2021, **Oral**) (Project site)

WORK EXPERIENCE

Research Scientist Intern, *Adobe Research, San Francisco* May. 2024 -
- Topics: video-guided Foley generation
- Supervisor: Prem Seetharaman, Bryan Russell, Justin Salamon

Research Scientist Intern, *Meta Reality Lab, Pittsburgh* May. 2023 - Nov. 2023
- Topics: neural acoustic fields

- Supervisor: Israel D. Gebru, Christian Richardt, Alexander Richard

Research Assistant, ***University of Michigan, Ann Arbor*** Jan. 2020 - Present

- Topics: multi-modal learning, self-supervised learning

Teaching Assistant, ***Shanghai Jiao Tong University*** Sept. 2018 - Aug. 2019

- Courses: *Digital Integrated Circuits*, *Electronic Circuits*
- Conducted experiments, held office hours and taught recitation classes every week

Research Engineer Intern, ***FreeSense Image Inc., Shanghai*** Mar. 2018 - May 2018

- Implemented an auxiliary system for a project of Detection of Car LCD Screen and applied it into industrial production lines
- Adopted state-of-the-art recognition DNNs to improve the accuracy on defect detection tasks

SERVICE

Reviewer

- Conference: WACV 2023, CVPR 2023, ICCV 2023, CVPR 2024, SIGGRAPH 2024, ECCV 2024, NeurIPS 2024
- Journal: IJCV

AWARDS & SCHOLARSHIPS

- **Lums Fellowship**, *University of Michigan, Ann Arbor* 2019-2021
- **Outstanding Graduate - Class of 2019**, *Shanghai Jiao Tong University* 2019
- **Academic Excellence Scholarship**, *Shanghai Jiao Tong University* 2016-2018
- **Dean's List**, *Shanghai Jiao Tong University, SJTU-UM JI* 2015-2018